

Supplementary Material: Semantic Identity Compression

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March 28, 2026

1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Handle	ID	Handle
ACS1	adversary_forces_n_minus_1_queries AbstractClassSystem/Extended.lean	ACS5	model_completeness AbstractClassSystem/Extended.lean
ACS7	nominal_localization_constant_semantic AbstractClassSystem/Extended.lean	ACS8	shape_cannot_distinguish AbstractClassSystem/Core.lean
ACS9	shape_provenance_impossible AbstractClassSystem/Core.lean	EMB1	semantic_non_embeddability AbstractClassSystem/Typing.lean
FRD1	ObserverModel.feasible_subsetMass_le_ optimalFeasibleMass Paper1IT/FiberRateDistortion.lean	FRD2	ObserverModel.optimalSubset_attains_ optimalFeasibleMass Paper1IT/FiberRateDistortion.lean
FXI1	fixed_axis_incompleteness axis_framework.lean	GPH13	Ssot.GraphEntropy.tagFeasible_iff_maxFiberCard_le
GPH16	Ssot.GraphEntropy.block_tagFeasible_iff_pow_ maxFiberCard_le Paper1IT/GraphEntropyAsymptotic.lean	GPH17	Ssot.GraphEntropy.maxFiberCard_blockObserve_eq_pow Paper1IT/GraphEntropyAsymptotic.lean
GPH18	Ssot.GraphEntropy.blockTagRateBits_eq_mul_oneShot Paper1IT/GraphEntropyAsymptotic.lean	GPH19	Ssot.GraphEntropy.blockTagRateBitsPerCoordinate_eq_ oneShot Paper1IT/GraphEntropyAsymptotic.lean
GPH20	Ssot.GraphEntropy.minBlockFeasibleAlphabet_eq_pow Paper1IT/GraphEntropyAsymptotic.lean	GPH24	Ssot.GraphEntropy.feasibleAtAlphabetBase_iff Paper1IT/GraphEntropyAsymptotic.lean
GPH25	Ssot.GraphEntropy.maxFiberCard_le_one_of_injective	GPH27	Ssot.GraphEntropy.tagFeasible_one_iff_injective
GPH28	Ssot.GraphEntropy.maxFiberCard_mono_of_factors_ through	GPH30	Ssot.GraphEntropy.uniformFiberErrorRate_ge_one_sub_ tag_ratio
GPH32	Ssot.GraphEntropy.optimalFiberBitLength_feasible	GPH33	Ssot.GraphEntropy.optimalExpectedAdaptiveBitLength_le
GPH34	Ssot.GraphEntropy.conditionalEntropyGiven_le_log2_ mul_expectedAdaptiveBitLength	GPH35	Ssot.GraphEntropy.exists_conditionalCodes_ expectedLength_le_entropy_bits_plus_one
GPH37	Ssot.GraphEntropy.optimalFiberBitLength_mono_of_ factors_through	GPH38	Ssot.GraphEntropy.taskRecoverable_iff_ maxTaskFiberCard_le_one
GPH39	Ssot.GraphEntropy.maxTaskFiberCard_mono_of_factors_ through	GPH40	Ssot.GraphEntropy.taskRecoverable_pair_iff
GPH41	Ssot.GraphEntropy.maxTaskFiberCard_pair_le_left	GPH42	Ssot.GraphEntropy.observeFiber_prod_card
GPH43	Ssot.GraphEntropy.maxFiberCard_prod	GPH44	Ssot.GraphEntropy.maxFiberCard_eq_finiteSup Paper1IT/ComputabilityQuantization.lean
GPH45	Ssot.GraphEntropy.maxFiberCard_computable_by_fiber_ enumeration Paper1IT/ComputabilityQuantization.lean	GPH48	Ssot.GraphEntropy.quantization_lower_bound_fin Paper1IT/ComputabilityQuantization.lean
GPH50	Ssot.GraphEntropy.precision_requirement Paper1IT/ComputabilityQuantization.lean	GPH52	Ssot.GraphEntropy.fiber_is_clique
GPH53	Ssot.GraphEntropy.fiber_finset_is_clique	GPH54	Ssot.QueryBitBridge.card_distinguished_set_le_two_ pow_card Paper1IT/QueryBitBridge.lean
GPH55	Ssot.QueryBitBridge.clog_card_le_query_count Paper1IT/QueryBitBridge.lean	GPH56	Ssot.QueryBitBridge.fiber_card_le_two_pow_query_ count Paper1IT/QueryBitBridge.lean
GPH57	Ssot.QueryBitBridge.maxFiberCard_clog_le_query_count Paper1IT/QueryBitBridge.lean	GPH60	Ssot.QueryBitBridge.distinguished_set_eq_full_ transcript_capacity Paper1IT/QueryBitBridge.lean
GPH61	Ssot.QueryBitBridge.clog_query_floor_is_tight_of_full_ capacity Paper1IT/QueryBitBridge.lean	GPH62	Ssot.QueryBitBridge.queryExpressivityGap_nonneg Paper1IT/QueryBitBridge.lean
GPH63	Ssot.QueryBitBridge.queryExpressivityGap_eq_zero_iff Paper1IT/QueryBitBridge.lean	GPH64	Ssot.QueryBitBridge.queryExpressivityGap_mono Paper1IT/QueryBitBridge.lean
INF1	AbstractClassSystem.Scope.observer_factors AbstractClassSystem/Extended.lean	L1	matroid_basis_equicardinality axis_framework.lean
L4	l4_exchange_wrapper HandleAliases.lean	L5	l5_exchange_wrapper HandleAliases.lean
L6	nonorthogonal_complete_has_redundant_axis axis_framework.lean	L7	exists_semanticallyMinimal_subset axis_framework.lean
L8	exists_orthogonal_semanticallyMinimal_subset axis_framework.lean	LWDC1	LWDCConverse.collision_block_requires_bits lwd_converse.lean
LWDC3	LWDCConverse.maximal_barrier_requires_bits lwd_converse.lean	NOH1	admissibility_no_hidden_state discipline_migration.lean
NSL1	AbstractClassSystem.Neurosymbolic.neurosymbolic_zero_ error_identity AbstractClassSystem/Neurosymbolic.lean	NSL2	AbstractClassSystem.Neurosymbolic.shape_alone_not_ zero_error AbstractClassSystem/Neurosymbolic.lean
NSL3	AbstractClassSystem.Neurosymbolic.neurosymbolic_ necessary_for_zero_error AbstractClassSystem/Neurosymbolic.lean	NSL4	AbstractClassSystem.Neurosymbolic.neurosymbolic_ identity_recovery AbstractClassSystem/Neurosymbolic.lean

ID	Handle	ID	Handle
NSL5	AbstractClassSystem.Neuropsychological.neuropsychological__ injective__implies__recoverable AbstractClassSystem/Neuropsychological.lean	NSL6	AbstractClassSystem.Neuropsychological.neuropsychological__is__ canonical__helper__view AbstractClassSystem/Neuropsychological.lean
PD1	nominal__pareto__dominates__shape discipline_migration.lean	PRIV3	AbstractClassSystem.Identity__disclosure__separation AbstractClassSystem/Extended.lean
PRV1	AbstractClassSystem.provenance__impossibility__universal	RDC1	ObserverModel.finiteRateDistortionConverse
RDC2	AbstractClassSystem/Extended.lean ObserverModel.finiteConditionalRateDistortionConverse	RDC3	Paper1IT/FiniteRateDistortionConverse.lean ObserverModel.finiteObservationOnlyRateDistortionConverse
RDC4	Paper1IT/FiniteRateDistortionConverse.lean ObserverModel.finiteMinEntropyBudgetConverse	RDC5	Paper1IT/FiniteRateDistortionConverse.lean ObserverModel.uniformFiniteRateDistortionConverse
RDC6	Paper1IT/FiniteRateDistortionBound.lean ObserverModel.finiteRateDistortionBound	RDC7	Paper1IT/FiniteRateDistortionConverse.lean ObserverModel.logBudgetLowerBoundFromError
RDC8	Paper1IT/FiniteRateDistortionBounds.lean ObserverModel.observationOnlyRateDistortionConverse	RDC9	Paper1IT/FiniteRateDistortionBounds.lean ObserverModel.observationOnlyMinEntropyBound
ROB1	Paper1IT/RateDistortion.lean AbstractClassSystem.OpenWorld.extend_to_force_barrier AbstractClassSystem/Undecidability.lean	ROB2	Paper1IT/RateDistortion.lean AbstractClassSystem.OpenWorld.barrierFreedom_not__ extension_stable AbstractClassSystem/Undecidability.lean
UND1	AbstractClassSystem.OpenWorld.hasBarrier_not__ computable AbstractClassSystem/Undecidability.lean	UND2	AbstractClassSystem.OpenWorld.barrierFree_not__ computable AbstractClassSystem/Undecidability.lean
ZDT1	Ssot.Paper1IT.requiredTagBits__le__iff__le__pow Paper1IT/GrowthTagBudget.lean	ZEC1	Ssot.GraphEntropy.zeroErrorConditionalEntropySandwich Paper1IT/ZeroErrorConditionalEntropy.lean